The Fine-Structure Constant as an OPH Pixel Fixed Point

Bernhard Müller Alexander Osika Daniel Ari Friedman

November 13, 2026

Paper release: r2066 **Released:** November 13, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

Abstract

A pixel fixed point constrains an electromagnetic coupling only when its return map is the response of the same physical system. We state this conditional identification, its scale and heat equations, and the same-scheme spectral transport needed at the Thomson endpoint. The common native-current and memory-mass construction has two certified conditional roots for a declared charged-fermion kernel: $P \simeq 1.630971514159316$, $\alpha^{-1} \simeq 137.000994766342$ with a supersymmetric running proxy, and $P \simeq 1.628533369611833$, $\alpha^{-1} \simeq 168.815082928453$ with the minimal census and color-scale coefficient 7. Neither is a complete electromagnetic response. Explicit phenomenological transport maps give certified inverse couplings 136.994835177413 . . . and 137.035660136947 . . .; replacing their quark-screening ansatz by measured hadronic data gives a value near 136.3827548, excluding the measured endpoint. Exact register counterexamples show why coarse counts cannot fix the missing spectral term. An independent entangled-pair operator supplies another exact scalar, but no Thomson identification. The resulting theorem requires a common current, physical same-cell map, consistent matter thresholds and a contraction certificate. Numerical agreement alone supplies none of those premises.

Keywords: fine-structure constant, Observer Patch Holography, pixel fixed point, electromagnetic coupling, Standard Model constants, hadronic vacuum polarization

1 The conditional response problem

A pixel closure identifies two readings only after a physical map is supplied: a geometric coordinate $(P - \varphi)/\sqrt{\pi}$ and the Thomson-limit response $1/A_{\mathrm{Th}}(P)$ of the same electromagnetic current. The carrier and gauge-response premises constrain the current algebra [1]; they do not determine its two-point function, matter thresholds or finite matching terms. A chosen simulator design is an admissible mathematical input. A source-only prediction additionally needs its parameter ancestry and independent physical validation.

The main executable connection uses the common current and charged-lepton mass operator of the particle construction [2]. Its charged-fermion feedback kernel, with the retained memory

mass law, gives two certified conditional roots (Section 10): $P \simeq 1.630971514159316$ and $\alpha^{-1} \simeq 137.000994766342$ with the supersymmetric running proxy and coefficient 4, whose extra fields are absent from the native matter carrier, and $P \simeq 1.628533369611833$ and $\alpha^{-1} \simeq 168.815082928453$ with the minimal three-family census and chosen color transmutation coefficient 7. With the same minimal census, coefficient 4, positive couplings, $P \geq 1$, $M_U \geq m_Z$ and no intervening thresholds, the heat/scale equations have no solution. Both memory inverse branches are certified at each root. These statements close a finite design consistency question; they identify no physical Thomson coupling. The count assumes the intended chiral continuum multiplicities, and the response omits charged-boson terms, confinement and full threshold matching. The localized interacting poles also differ from the quadratic coefficients used by this loop.

The transport maps specified below isolate the separate endpoint problem. They use a supplied phenomenological charged spectrum, the MSSM running coefficients and a declared quark-screening ansatz. Their certified inverse couplings are 136.994835177413 . . . and 137.035660136947 . . .; neither is the measured endpoint 137.035999177(21). Replacing the screening ansatz by the measured hadronic payload gives 136.3827548, outside the measured interval. These are explicit test kernels, not complete electromagnetic response functions of the native simulator. Their certificates verify the specified functions and do not bound omitted physical contributions. The hadronic comparison uses the standard dispersion framework [11, 12, 15]; the exact independent pair construction appears in Section 12.

Proposition 1 (Maxwell normalization of the endpoint branch)**.** *On the declared electromagnetic branch used by the fixed-point equation, the Ward-projected* $\mathrm{U}(1)_Q$ *channel carries*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2;P){*}J_Q,$$

with

$$g_Q^2(q^2;P) = 4\pi\alpha_{\mathrm{em}}(q^2;P), \qquad A_{\mathrm{Th}}(P) = \alpha_{\mathrm{em}}^{-1}(0;P).$$

Thus the inside reading in the pixel equation is the Thomson-limit Maxwell coupling of the same electromagnetic current.

Proof. On the declared unbroken Maxwell action of the gauge paper, the electromagnetic branch is the abelian factor $\mathrm{U}(1)_Q$. There the compact-gauge curvature reduces to $F_Q = dA_Q$, and the quadratic field action gives $d{*}F_Q = g_Q^2{*}J_Q$. That action is a branch premise: the executable observer supplies finite Maxwell-shaped seam actions, and a local source-produced physical Maxwell theory on Lorentzian spacetime is not constructed [6]. The declared Ward-current transport premise identifies the same $g_Q(q^2;P)$ with the Ward-projected running electromagnetic coupling. Taking $q^2 \to 0$ gives the endpoint quantity $A_{\mathrm{Th}}(P)$ used in the pixel fixed-point equation. □

The hadronic step uses the standard dispersion logic behind hadronic vacuum-polarization work. The relevant comparison data are electromagnetic spectral functions measured through $e^+e^- \to$ hadrons, as used in data-driven running-α and $g-2$ analyses [11, 12, 15]. In this paper, the decimal correction shown above is reported as the difference between the CODATA/NIST comparison value and the OPH declared-kernel calculation. The source data in this paper emit no direct production calculation of the hadronic spectral function; a pure source endpoint theorem consumes that calculation as a premise.

2 Symbols

Symbol	Definition	Informal meaning
$\mathcal{A}(P)$	Patch algebra assigned to a screen patch P	The local observables available to one finite observer patch.
ω_P	State on $\mathcal{A}(P)$	The local physical description carried by that patch.
$\mathcal{A}(P \cap Q)$	Shared overlap algebra	The observables two neighboring patches can compare.
P	$a_{\mathrm{cell}}/\ell_\star^2$	The dimensionless area of one screen cell in scale-readout units.
a_{cell}	Declared screen-cell area	The area assigned to the local pixel by the declared screen dictionary.
$\ell_\star$	OPH scale-certificate length	The length emitted by the selected no-G scale certificate.
ℓ_P	Planck length	The SI display of $\ell_\star$ after $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$.
φ	$(1+\sqrt{5})/2$	The self-similar entropy-balance point.
$\sqrt{\pi}$	Boundary Gaussian normalization width	The conversion factor from pixel displacement to observation width.
$\alpha_{\mathrm{ext}}(P)$	$(P-\varphi)/\sqrt{\pi}$	The declared outer detuning coordinate.
$A_{\mathrm{Th}}(P)$	$\alpha_{\mathrm{em}}^{-1}(0;P)$	The inverse electromagnetic endpoint returned by the declared trial map.
$\alpha_{\mathrm{in}}(P)$	$1/A_{\mathrm{Th}}(P)$	The declared inner coordinate; its same-cell interpretation requires a typed bridge.
E_P	Planck energy	The energy unit associated with the selected scale certificate. The numerical equations use Planck units unless GeV labels are attached.
$M_U(P)$	$E_P \mathrm{e}^{-2\pi} P^{1/6}$	The OPH unification-scale readout returned at the declared trial coordinate.
$E_{\mathrm{cell}}(P)$	$E_P/\sqrt{P}$	The local cell energy readout.
$\alpha_U(P)$	Unified coupling solved from the heat-kernel closure equation	The coupling at $M_U(P)$ that makes the gauge representation entropy match the pixel.
$\alpha_i(\mu;P)$	Gauge coupling $i=1,2,3$ at scale μ	The hypercharge, weak, and color couplings emitted by the same trial P.

Symbol	Definition	Informal meaning
b_i	$(33/5, 1, -3)$	The one-loop MSSM beta coefficients used as the high-scale running convention here; a declared structural selection.
$m_Z(P)$	Self-consistent electroweak anchor scale	The Z-scale generated by $v(P)$, $\alpha_1(P)$, and $\alpha_2(P)$.
$v(P)$	$E_{\mathrm{cell}}(P)\exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U(P))]$	The electroweak transmutation scale.
β_{EW}	$N_c + 1 = 4$ for $N_c = 3$	The coefficient that controls the exponential drop to the electroweak scale; the inverse plaquette cost of one weak-centre flux.
$A_Z(P)$	$\alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$	The electroweak-scale electromagnetic anchor.
$\Delta_{\mathrm{lep}}(P)$	Lepton transport contribution to inverse alpha	The exact one-loop kernel evaluated on the frozen phenomenological e, μ, τ spectrum.
$\Delta_{\mathrm{had}}(P)$	Hadronic transport contribution	The low-energy QCD spectral contribution needed for the declared-kernel calculation.
$\Delta_{\mathrm{EW}}(P)$	Electroweak/scheme endpoint remainder	Same-scheme finite remainder needed to connect the source anchor to the endpoint.
$\Delta_{\mathrm{calc}}(P)$	Calculated charged-fermion transport contribution	The exact kernel on the supplied lepton continuation plus the screened quark continuation; not a pure source prediction.
$\Delta_{\mathrm{H,cal}}^{\mathrm{fp}}$	Root-to-CODATA bookkeeping difference	The inverse-alpha amount between the naked root value and the CODATA/NIST central endpoint; it includes the comparison-pixel $\alpha_U(P_{\mathrm{C}})$ contribution and is not the post-α_U hadronic gap.
$\Delta_{\mathrm{H,req}}^{\mathrm{fp}}$	Required hadronic endpoint correction after $A_{\alpha_U}^{\mathrm{fp}}$	The inverse-alpha amount by which the displayed OPH output misses the CODATA/NIST central endpoint.
$R_Q(P)$	$A_{\mathrm{Th}}(P) - [A_Z(P) + \Delta_{\mathrm{calc}}(P)]$	The residual same-scheme hadronic endpoint contribution.
$A_{\alpha_U}^{\mathrm{fp}}$	$\alpha_{\mathrm{root}}^{-1} + \alpha_U(P_{\mathrm{C}})$	Mixed-provenance no-hadron comparison diagnostic; the root is source-side and P_{C} is CODATA-derived.

Symbol	Definition	Informal meaning
$J_{24,Q}(P)$, $\omega_Q(P)$, $\Xi_Q(P)$	Source-side hadronic spectral objects required for a final endpoint theorem	The Jacobi/Stieltjes spectral payload, normalization, and finite same-scheme remainder that must be emitted without looking at the Thomson target.
P_{C}	CODATA/NIST comparison pixel	The pixel obtained from the CODATA/NIST Thomson endpoint under the OPH outer equation.

3 Shared Collar Convention for χ_ν

The χ_ν protected-reserve theorem uses the alpha-calibrated comparison pixel

$$P_\chi = P_{\mathrm{C}} = 1.630968209403959\ldots.$$

Its scalar reserve density is

$$\epsilon_{\mathrm{res}} = \frac{P_\chi}{24}$$

only after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass in the screen microphysics paper and in the observers library paper. This does not make $P/4$ a primitive Hilbert-space dimension. In this paper, $P/4$ is the local screen-cell entropy budget used by the electromagnetic fixed-point branch; $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch. The $P/4$ budget is a declared area reading of the cell. Native regional entropy in the executable observer grows with the number of enclosed ports rather than with crossing seams, and the physical region map that would carry an area law is a premise [9].

4 The pixel coordinate and return condition

4.1 The OPH Overlap Rule

OPH starts from a finite screen covered by observer patches. If P_1 and P_2 overlap, then their induced states must agree on the shared algebra:

$$\omega_{P_1}|_{\mathcal{A}(P_1\cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1\cap P_2)}. \tag{1}$$

The quantitative fine-structure branch applies this rule to a single local screen cell. The cell has one outside description and one inside description. The outside description is geometric. The inside description is electromagnetic. A declared typed bridge identifies both as readings of one cell; the fixed-point theorem is conditional on that premise.

The executable observer construction runs local agreement repair on a twelve-port carrier: each native repair event moves load across one seam and records its placement coin and the transferred current [10]. Its physical identification with our Universe is unestablished.

4.2 The Pixel Variable

Define the local pixel ratio

$$P := \frac{a_{\mathrm{cell}}}{\ell_\star^2}. \tag{2}$$

Here a_{cell} is the screen-cell area and $\ell_\star^2$ is the area scale emitted by the separate OPH gravity readback fixed point.

4.3 The Golden-Ratio Balance

The self-similar balance point is

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 - \varphi - 1 = 0, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{3}$$

The pixel does not sit exactly at φ. It sits at

$$\Delta_P := P - \varphi. \tag{4}$$

4.4 Boundary Gaussian Normalization

The boundary normalization converts Δ_P into the outside coupling readout:

$$\alpha_{\mathrm{ext}}(P) := \frac{P-\varphi}{\sqrt{\pi}}. \tag{5}$$

4.5 The Inside Electromagnetic Readout

Let

$$A_{\mathrm{Th}}(P) := \alpha_{\mathrm{em}}^{-1}(0;P) \tag{6}$$

be the inverse electromagnetic coupling at zero momentum returned by the declared map at trial pixel P. The inside coupling is

$$\alpha_{\mathrm{in}}(P) := \frac{1}{A_{\mathrm{Th}}(P)}. \tag{7}$$

4.6 The Fixed-Point Equation

Conditional on the same-invariant bridge, closure requires the outside and inside couplings to agree:

$$\alpha_{\mathrm{ext}}(P) = \alpha_{\mathrm{in}}(P). \tag{8}$$

Using Eqs. (5) and (7),

$$\frac{P-\varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P)}. \tag{9}$$

Equivalently,

$$\boxed{H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)} = 0} \tag{10}$$

or

$$\boxed{P = G(P) := \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)}.} \tag{11}$$

5 Scale, running and heat closure

5.1 The Source Map

For a trial P, the source map first emits

$$M_U(P) = E_P \, \mathrm{e}^{-2\pi} \, P^{1/6}, \tag{12}$$

$$E_{\mathrm{cell}}(P) = \frac{E_P}{\sqrt{P}}. \tag{13}$$

Eq. (13) is one radian of phase per tick when a screen cell is the face dual to one link and a tick moves a massless quantum by one link: the link has length $\sqrt{a_{\mathrm{cell}}}$, the tick lasts $\sqrt{a_{\mathrm{cell}}}/c$, and $\hbar c/\sqrt{a_{\mathrm{cell}}} = E_P/\sqrt{P}$.

5.2 Electroweak Transmutation

The source branch uses

$$\beta_{\mathrm{EW}} = N_c + 1 = 4 \tag{14}$$

with $N_c = 3$. The same value has a representation-theoretic reading in the Standard Model gauge paper: the selected exterior matter package contains exactly four weak doublets, $3 + 1$ from the quark and lepton sectors of one generation, so $N_c + 1$ and the weak-doublet multiplicity name one coefficient. The identification of that abstract load with the physical screen load is the separate common-carrier receipt. The branch defines

$$v(P, \alpha_U) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\mathrm{EW}} \alpha_U}\right]. \tag{15}$$

The same integer is the inverse plaquette cost of the weak centre. The central element $\mathrm{diag}(1, 1, 1, -1, -1)$ of the code group is the class-three flux of the protected defects and the 2π rotation of the carrier's cover. Its shortest generator is $H_3 = \mathrm{diag}(0, 0, 0, i\pi, -i\pi) = 2\pi i \, T_3$, with $\mathrm{tr}_5 \, H_3^\dagger H_3 = 2\pi^2$. For $\alpha_U = g^2/4\pi$ the exponent of Eq. (15) is the Gaussian plaquette action of one weak-centre flux quantum,

$$\frac{2\pi}{\beta_{\mathrm{EW}} \alpha_U} = \frac{\mathrm{tr}_5 \, H_3^\dagger H_3}{g^2}, \tag{16}$$

and the shortest generators of the five nontrivial central classes give

$$\frac{8\pi^2}{\mathrm{tr}_5 \, H_k^\dagger H_k} = \frac{12}{5}, \; 3, \; 4, \; 3, \; \frac{12}{5} \qquad (k = 1, \ldots, 5) \tag{17}$$

[1, 2]. The weak-centre reading fixes the integer; the prefactor E_{cell} and the identification of the scale with the electroweak vacuum are the premises of Eq. (15).

5.3 One-Loop Gauge Running

For $i = 1, 2, 3$, define

$$(b_1, b_2, b_3) = \left(\frac{33}{5},\ 1,\ -3\right), \tag{18}$$

and run the couplings by

$$\alpha_i^{-1}(\mu; P, \alpha_U) = \alpha_U^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right). \tag{19}$$

The triple $(33/5, 1, -3)$ is the one-loop MSSM beta-coefficient set. Its census requires superpartners beyond the native fifteen-state matter carrier. No supersymmetric threshold matching enters this map, so it is a declared running proxy rather than the response of that carrier. The minimal-census exclusion and the color-scale closure are in Section 10.

The executable source has its own one-loop content. Its 32 matter patterns per copy form $16 + \overline{16}$ in three copies, with twelve gauge currents and n_H complex Higgs doublets, and the coefficients are

$$(b_1, b_2, b_3) = \left(8 + \frac{n_H}{10},\ \frac{2}{3} + \frac{n_H}{6},\ -3\right).$$

For $n_H = 2$ the weak and colour coefficients equal the declared $(1, -3)$ with no superpartner, and the hypercharge coefficient is $41/5$ against the declared $33/5$. The source as built has one doublet, $(81/10, 5/6, -3)$. Because the $\overline{16}$ half is a sum of complete multiplets, the slope differences are those of the Standard Model, and equal couplings at M_U give $(1/\alpha_2 - 1/\alpha_3)/(1/\alpha_1 - 1/\alpha_2) = 115/218$ at m_Z; the declared slopes give $5/7$ and the measured couplings 0.717. On the chain the source as built gives $v = 222.9$ GeV, $\alpha_s(m_Z) = 0.1123$ and $\sin^2\theta_W(m_Z) = 0.2056$. The hypercharge slope of the declared chain has no source reading.

5.4 The Self-Consistent Z-Scale

The hypercharge coupling in Standard Model normalization is

$$\alpha_Y(\mu; P) = \frac{3}{5}\alpha_1(\mu; P). \tag{20}$$

The tree-level Z-mass readout is

$$m_Z(\mu; P, \alpha_U) = \frac{v(P, \alpha_U)}{2}\sqrt{4\pi\alpha_2(\mu; P, \alpha_U) + 4\pi\alpha_Y(\mu; P, \alpha_U)}. \tag{21}$$

The source point uses the self-consistency condition

$$\mu = m_Z(\mu; P, \alpha_U). \tag{22}$$

5.5 Heat-Kernel Gauge Closure

For SU(2), the irreducible representations are labeled by $j = n/2$, $n = 0, 1, \ldots, N_2$, with

$$d_j = 2j + 1, \qquad C_2(j) = j(j+1). \tag{23}$$

For SU(3), representations are labeled by highest weights (p, q), $0 \le p, q \le N_3$, with

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}, \tag{24}$$

$$C_2(p,q) = \frac{p^2+q^2+pq+3p+3q}{3}. \tag{25}$$

For a compact group G in this finite representation cutoff, define

$$Z_G(t) = \sum_R d_R \, \mathrm{e}^{-tC_2(R)}, \tag{26}$$

$$\bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R \, \mathrm{e}^{-tC_2(R)} \log d_R. \tag{27}$$

The displayed evaluations use the representation cutoffs $N_2 = 120$ and $N_3 = 90$. At the closure arguments below, $t_2 \approx 1.3$ and $t_3 \approx 4.7$, the first omitted terms carry weights below e^{-4000}, so the cutoffs are inert at every displayed precision, and the interval certificate bounds the tails beyond any finite cutoff.

The heat-kernel source parameters are

$$t_2 = 4\pi^2 \alpha_2(m_Z; P, \alpha_U), \qquad t_3 = 4\pi^2 \alpha_3(m_Z; P, \alpha_U). \tag{28}$$

In the modular reading of the collar the factor $4\pi^2$ is the modular period times the electric normalization: a sector R with electric energy $(g^2/2)C_2(R)$ at unit distance from the cut has the modular weight $\exp[-2\pi(g^2/2)C_2(R)] = \exp[-4\pi^2 \alpha \, C_2(R)]$. The product of modular distance and electric energy does not depend on the size of the collar, so the scale enters only through the running couplings. The pixel-closure equation is

$$\boxed{\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) = \frac{P}{4}.} \tag{29}$$

For a trial P, Eq. (29) is solved for $\alpha_U(P)$. The closure sums the SU(2) and SU(3) representation towers only. Hypercharge carries no representation entropy: every irreducible representation of U(1) has dimension one, so $\log d_R = 0$ and $\bar{\ell}_{\mathrm{U}(1)}(t) = 0$ for every t. The repair round count used by the finite current construction counts the $\mathfrak{u}(1)$ generator $(8 + 3 + 1 = 12)$; the two counts measure different quantities.

The couplings in Eq. (28) are read at the first freeze of the weak currents. A Higgs doublet gives the two charged directions the squared mass $\pi\alpha_2 v^2$ and the neutral direction $\pi(\alpha_2 + \alpha_Y)v^2$, so the neutral direction is the heaviest and freezes first, at the fixed point of Eq. (22). Above that scale the three weak directions enter the collar generator with one coefficient, which is the hypothesis of the heat-kernel law $d_R \, \mathrm{e}^{-tC_2(R)}$; below it the surviving generator separates the two members of a doublet. The colour factor of the same collar is read at the same scale. The reading has one premise: below the anchor the remaining transport and charge terms act on the fused label of a collar and on no factor of a single cell.

5.6 The Electroweak Anchor

Once $\alpha_U(P)$ and $m_Z(P)$ are solved, define

$$\alpha_Y(m_Z;P) = \frac{3}{5}\alpha_1(m_Z;P), \tag{30}$$

$$\alpha_{\rm em}(m_Z^2;P) = \left(\frac{1}{\alpha_2(m_Z;P)} + \frac{1}{\alpha_Y(m_Z;P)}\right)^{-1}. \tag{31}$$

The source-locked electroweak anchor is

$$\boxed{A_Z(P) := \alpha_{\rm em}^{-1}(m_Z^2;P).} \tag{32}$$

The weak mixing readout is

$$\sin^2\theta_W(m_Z;P) = \frac{\alpha_{\rm em}(m_Z^2;P)}{\alpha_2(m_Z;P)}. \tag{33}$$

5.7 Outputs of the chain and the standing of its parts

At the fixed point $P = 1.6309721$ the chain gives $1/\alpha_U = 24.3166$ at $M_U = 2.4736 \times 10^{16}$ GeV, $v = 246.60$ GeV, tree masses $m_Z = 91.588$ and $m_W = 80.330$ GeV, $1/\alpha_2(m_Z) = 29.605$, $\alpha_s(m_Z) = 0.11834$, $\sin^2\theta_W(m_Z) = 0.23074$ and $1/\alpha_{\rm em}(m_Z) = 128.308$. The measured values are 246.22, 91.188 and 80.369 GeV, 29.58, 0.1180(9), 0.23122 and 127.95. With the measured couplings at m_Z the left side of the closure is 0.407641 against $P/4 = 0.407743$.

Four parts of the chain have a source reading: the cell energy, the count 4 of the transmutation law as the plaquette cost of one weak-centre flux quantum, the anchor at the first freeze of the weak currents under one premise on the collar generator below it, and the factors 2π as the modular period. Three parts are declared. The unification scale of the stack itself is $E_{\rm cell}e^{-2\pi} = 1.7853\times10^{16}$ GeV, which gives $v = 264.6$ GeV; the further factor $P^{2/3}$ of Eq. (12) has no source reading. The collar law's weight d_R and unit shape number are the only pair that closes the chain: weight d_R^2 gives $1/\alpha_U = 16.71$, and a shape number of one half gives 11.44. The chain is consistent only for couplings that run exactly at one loop; in the $\overline{\rm MS}$ scheme at two loops the weak and colour couplings of the source content meet at the same scale within 0.013 in the logarithm and at $1/\alpha = 22.67$. The transmutation law has the structure $v = E_{\rm cell}\, b\, \langle A\rangle$ with $\langle A\rangle$ the class balance of one seam link; its exponent needs a link whose plaquette cannot move, and lattice ensembles give a coefficient between 0.33 and 0.50 in place of one. The chain is declared and its outputs are compared after the measured values were known.

6 Charged input and electromagnetic transport

6.1 Frozen Charged-Spectrum Continuation Used by the Transport

The one-loop kernel below is exact for supplied fermion masses, but the charged spectrum supplied to it in this section is a phenomenological continuation, not an OPH-derived mass theorem. It uses $N_c = 3$, $N_g = 3$, and the additional assumptions

$$\epsilon = \frac{1}{6}, \qquad \delta = \frac{\beta_{\rm EW}}{2N_cN_g} = \frac{2}{9}. \tag{34}$$

No physical charged-spectrum derivation is claimed here; such a derivation requires a source-to-mass attachment, a phase-transport law fixing $\delta = 2/9$, and an attachment of the ordered roots to physical charged-family lines. For a positive-semidefinite square-root-mass carrier

$$C = aI + \rho(e^{i\delta}R + e^{-i\delta}R^2), \qquad Q = \frac{1 + 2(\rho/a)^2}{3},$$

so $\rho/a = 1/\sqrt{2}$ is exactly equivalent to the empirical Koide value $Q = 2/3$. At balance this identity holds throughout the positive chamber $|\delta| \le \pi/12 \pmod{2\pi/3}$, and it does not select the phase. The balance is a target-informed premise and the phase δ is supplied. The finite tracial construction that realizes the balance, and the maps a physical charged-lepton reading would additionally require, are stated in the Koide paper [4]. Define three Koide roots

$$r_k = 1 + \sqrt{2}\cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2, \tag{35}$$

then sort them in increasing order and write the sorted list as (r_1, r_2, r_3).

The quark exponent vectors are

$$\boldsymbol{n}_u = (2N_c, N_c, 0) = (6, 3, 0), \qquad \boldsymbol{n}_d = (2N_c, N_c + 1, N_c - 1) = (6, 4, 2). \tag{36}$$

With $v = v(P)$,

$$m_u = \frac{v}{\sqrt{2}}\epsilon^6, \qquad m_c = \frac{v}{\sqrt{2}}\epsilon^3, \qquad m_t = \frac{v}{\sqrt{2}}, \tag{37}$$

$$m_d = \frac{v}{\sqrt{2}}\epsilon^6, \qquad m_s = \frac{v}{\sqrt{2}}\epsilon^4, \qquad m_b = \frac{v}{\sqrt{2}}\epsilon^2. \tag{38}$$

For charged leptons the exponent vector is

$$\boldsymbol{n}_e = (7, 4, 3). \tag{39}$$

This vector and the normalization below are declared phenomenological transport inputs. The declared premises do not derive $\boldsymbol{n}_e = (7, 4, 3)$ or the resulting determinant identity $\det M_e = v(P)^3/(2\,6^{14})$. The electron entry is selected by a finite scan: with $n_\tau = N_g = 3$ and $n_\mu = N_g + 1 = 4$ fixed by the generation count, the electron exponent runs over the integers 5 to 11, and the entry with the smallest maximal pairwise logarithmic residual between the squared balanced-circulant roots at $\delta = (N_c + 1)/(2N_cN_g)$ and the corresponding powers of $\epsilon = 1/6$ is kept; the minimizer is 7. The scan compares the declared circulant roots with the ϵ ladder and reads no measured mass, and the vector is a declared transport input all the same. Every common shift $(7 + k, 4 + k, 3 + k)$ has the same ratio residual while changing the determinant. The extra factor $2^{1/6}$ below is asserted rather than derived. Define

$$\log g_c = \frac{1}{3}\sum_{a=1}^{3} \log\left(\frac{r_a^2\sqrt{2}\,6^{n_{e,a}}}{v}\right), \qquad s_0 = \mathrm{e}^{-\log g_c}, \qquad s_e = s_0\,2^{1/6}. \tag{40}$$

Then

$$m_e = s_e r_1^2, \qquad m_\mu = s_e r_2^2, \qquad m_\tau = s_e r_3^2. \tag{41}$$

Equations (40)–(41) define a transport input, not a source-derived charged-lepton mass law. No charged RG/threshold theorem maps this coordinate to physical pole masses.

6.2 Exact One-Loop Fermion Transport

For a fermion of mass m_f, electric charge Q_f, and multiplicity N_f, define

$$K_f(Q^2; m_f, Q_f, N_f) = \frac{2N_f Q_f^2}{\pi} \int_0^1 x(1-x) \log\left(1 + \frac{Q^2 x(1-x)}{m_f^2}\right) \mathrm{d}x. \tag{42}$$

The integral has the following closed form. Let

$$z = \frac{Q^2}{m_f^2}, \qquad a = \frac{z}{4}. \tag{43}$$

Then

$$\int_0^1 x(1-x) \log(1 + zx(1-x)) \,\mathrm{d}x = -\frac{5}{18} + \frac{1}{6a} + \frac{(2a-1)\sqrt{1+a}\,\operatorname{asinh}(\sqrt{a})}{6a^{3/2}}, \tag{44}$$

with

$$\operatorname{asinh}(\sqrt{a}) = \log(\sqrt{a} + \sqrt{1+a}). \tag{45}$$

The lepton contribution is

$$\Delta_{\mathrm{lep}}(P) = K_e(m_Z(P)^2; m_e, 1, 1) + K_\mu(m_Z(P)^2; m_\mu, 1, 1) + K_\tau(m_Z(P)^2; m_\tau, 1, 1). \tag{46}$$

The naive five-quark contribution is

$$\begin{aligned}\Delta_q^{\mathrm{naive}}(P) = {} & K_u(m_Z(P)^2; m_u, 2/3, 3) + K_d(m_Z(P)^2; m_d, -1/3, 3) \\ & + K_s(m_Z(P)^2; m_s, -1/3, 3) + K_c(m_Z(P)^2; m_c, 2/3, 3) + K_b(m_Z(P)^2; m_b, -1/3, 3).\end{aligned} \tag{47}$$

The declared hadronic screening ansatz is

$$S_{\mathrm{calc}}(P) = 1 - \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{48}$$

The form of S_{calc} is a declared selection of the quark continuation and carries no perturbative derivation: the leading perturbative correction to quark vacuum polarization is the enhancement $R \to R_0(1 + \alpha_s/\pi + \cdots)$ with no N_c factor, about $+3.7$ percent at m_Z, while $S_{\mathrm{calc}} \approx 0.887$ is an 11 percent suppression standing in for confinement cutting off the light-quark logarithms. Thus

$$\Delta_{\mathrm{calc}}(P) = \Delta_{\mathrm{lep}}(P) + S_{\mathrm{calc}}(P) \Delta_q^{\mathrm{naive}}(P). \tag{49}$$

7 Endpoint calculations and comparison

7.1 The Source Calculation Endpoint

The calculated endpoint is

$$A_{\mathrm{calc}}(P) = A_Z(P) + \Delta_{\mathrm{calc}}(P). \tag{50}$$

Putting A_{calc} into Eq. (11) gives the source map

$$G_{\mathrm{calc}}(P) = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{calc}}(P)}. \tag{51}$$

The numerical solve uses

$$G_{\mathrm{calc}}(P_{\mathrm{root}}) = P_{\mathrm{root}} \tag{52}$$

and emits

$$P_{\mathrm{root}} = 1.630972095858897376964513903506955628479\ldots, \tag{53}$$

$$\alpha_{\mathrm{root}}^{-1} = 136.994835177412937295289429464436902857 6\ldots. \tag{54}$$

The fixed point is interval-certified as existing and unique on its interval (interval certificate, available in the public artifact bundle [3]; enclosure widths 6.8×10^{-28} in P and 7.2×10^{-24} in α^{-1}, Lipschitz bound $L \leq 0.0724$). The source anchor at the certified root point is

$$A_Z(P_{\mathrm{root}}) = 128.308268057987597347904057614084758875033 1\ldots. \tag{55}$$

The calculated transport contribution is

$$\Delta_{\mathrm{calc}}(P_{\mathrm{root}}) = 8.686567119425339947385371850352143982 52\ldots. \tag{56}$$

The P-dependent detector of the executable observer construction takes this root as its closure input. Under a chosen charge assignment it sets $g(P_{\mathrm{root}}) = \sqrt{4\pi\alpha(P_{\mathrm{root}})} = 0.3028676131\ldots$ and records a binary detector observation whose probabilities depend on P and which feeds the next native repair; that use leaves the endpoint status of the root unchanged [8].

7.2 The CODATA/NIST Measured Endpoint and Comparison Pixel

The CODATA/NIST 2022 inverse fine-structure constant [17, 19] is

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{57}$$

with concise form 137.035999177(21). The corresponding comparison pixel is

$$P_{\mathrm{C}} = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{C}}} = 1.63096820940395932487\ldots \tag{58}$$

The CODATA/NIST central coupling is

$$\alpha(0) = \frac{1}{A_{\mathrm{C}}} = 0.00729735256433142503\ldots \tag{59}$$

Two-P provenance audit. Two distinct pixel numerals enter this transport comparison, and the feedback roots of Section 10 are separate from both. The declared-kernel solve gives P_{root} in Eq. (53); it is the branch that audits what the OPH source chain emits before a same-scheme hadronic endpoint correction is supplied. The comparison value P_{C} in Eq. (58) is different by about 3.9×10^{-6} in P-space because it is defined from the measured CODATA/NIST Thomson endpoint. Thus the published comparison pixel is a measured-endpoint display, not a derivation of α. Downstream quantities that use P_{C} inherit that measured endpoint by definition; downstream quantities that use P_{root} are source-branch diagnostics. The registered conditional primordial tilt $n_s = 1 - P/48 = 0.9660214956\ldots$ is one such downstream quantity; it is evaluated at P_{C} [5, 7].

7.3 Endpoint Accounting at the CODATA/NIST Comparison Pixel

At $P = P_{\mathrm{C}}$, the source point gives

$$A_Z(P_{\mathrm{C}}) = 128.30796547328624820996\ldots \tag{60}$$

$$\Delta_{\mathrm{calc}}(P_{\mathrm{C}}) = 8.68656784270852840098\ldots \tag{61}$$

These are abbreviated diagnostic outputs. The full decimal values and certified interval artifacts are supplied with the executable calculation; finite printed precision is not an additional certificate. The transport required by the CODATA/NIST measured endpoint is

$$\Delta_{\mathrm{req}}(P_{\mathrm{C}}) = A_{\mathrm{C}} - A_Z(P_{\mathrm{C}}) = 8.72803370371375179003\ldots \tag{62}$$

Therefore the source-side residual at the CODATA/NIST comparison pixel is

$$\begin{aligned} R_Q(P_{\mathrm{C}}) &= \Delta_{\mathrm{req}}(P_{\mathrm{C}}) - \Delta_{\mathrm{calc}}(P_{\mathrm{C}}) \\ &= 0.04146586100522338905\ldots \end{aligned} \tag{63}$$

7.4 The OPH Output and the Required Hadronic Correction

The fixed-point calculation emits

$$A_{\mathrm{calc}}(P_{\mathrm{root}}) = \alpha_{\mathrm{root}}^{-1} = 136.99483517741293729528\ldots. \tag{64}$$

The interval certificate and continuous-integration identity test are included in the public artifact bundle [3]. At the CODATA-derived comparison pixel, the finite-screen unified gauge-width contribution is

$$\alpha_U(P_{\mathrm{C}}) = 0.041124336195630495.$$

Combining that comparison-pixel gauge width with Equation (64) gives the mixed-provenance no-hadron diagnostic

$$A^{\mathrm{fp}}_{\alpha_U} = A_{\mathrm{calc}}(P_{\mathrm{root}}) + \alpha_U(P_{\mathrm{C}}) = 137.03595951360856779028942946443 7\ldots, \tag{65}$$

or

$$\alpha^{\mathrm{fp}}_{\mathrm{src}+U} = \left(A^{\mathrm{fp}}_{\alpha_U}\right)^{-1} = 0.00729735467646135213287275959657\ldots. \tag{66}$$

This $A^{\mathrm{fp}}_{\alpha_U}$ value is comparison bookkeeping, not a source-only prediction: $\alpha_{\mathrm{root}}^{-1}$ is evaluated on the source root, while $\alpha_U(P_{\mathrm{C}})$ inherits the measured endpoint through P_{C}, and the sum is not a fixed point of any single declared map.

Compared with the CODATA/NIST central measured endpoint,

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{67}$$

the CODATA-minus-diagnostic bookkeeping gap is

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = A_{\mathrm{C}} - A^{\mathrm{fp}}_{\alpha_U} = 0.00003966339143220971\ldots\ldots. \tag{68}$$

Equivalently,

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = 0.0000396634 \pm 0.000000021, \qquad \frac{\Delta^{\mathrm{fp}}_{\mathrm{H,req}}}{A_{\mathrm{C}}} = 2.89438 \times 10^{-7}. \tag{69}$$

This is the inverse-alpha gap between the mixed diagnostic and the CODATA/NIST central value. It is the gap left by the declared screening ansatz; inserting the measured hadronic vacuum polarization instead does not close it (Section 8.1). The subtraction neither calculates same-scheme transport of the Ward-projected electromagnetic current nor identifies the physical source of the gap. The 2.89438×10^{-7} ratio describes the mixed diagnostic only. The distinct self-consistent gauge-width map has fixed point $\alpha^{-1} = 137.035660136946577\ldots$ and relative residual 2.5×10^{-6}, about 1.6×10^{4} measurement sigma.

For bookkeeping, the full root-to-CODATA difference is

$$\begin{aligned} \Delta^{\mathrm{fp}}_{\mathrm{H,cal}} &= A_{\mathrm{C}} - A_{\mathrm{calc}}(P_{\mathrm{root}}) \\ &= 0.041163999587062704 71\ldots \\ &= \alpha_U(P_{\mathrm{C}}) + \Delta^{\mathrm{fp}}_{\mathrm{H,req}}. \end{aligned} \tag{70}$$

This larger difference is bookkeeping: it contains the comparison-pixel unified gauge-width contribution and the CODATA-minus-diagnostic gap.

Equivalently, the CODATA/NIST central value can be written as

$$A_{\mathrm{C}} = \alpha^{-1}_{\mathrm{root}} + \alpha_U(P_{\mathrm{C}})\, C_{24,Q}, \qquad C_{24,Q} = 1.000964474933857388040135711404515 2\ldots. \tag{71}$$

The factor $C_{24,Q}$ is a back-solved calibration: compact comparison accounting, solved from the measured endpoint, for the finite same-scheme hadronic endpoint transport needed for source-only closure. Equivalently, $\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = \alpha_U(P_{\mathrm{C}})(C_{24,Q} - 1)$. The factor is not a source-emitted hadronic/QCD payload.

Here A_{C} is the CODATA/NIST 2022 inverse fine-structure constant [17, 18]. Closing the endpoint gap requires a source-derived hadronic spectral calculation, which this paper does not supply.

The endpoint decomposition is

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P). \tag{72}$$

The calculated transport contains the exact one-loop charged-lepton contribution and a structured quark-screening continuation. The residual pure source object can be written compactly as

$$R_Q(P) = \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P) - [\Delta_{\mathrm{calc}}(P) - \Delta_{\mathrm{lep}}(P)], \tag{73}$$

where the right side is understood in the same renormalization and endpoint scheme as $A_Z(P)$.

In the screening notation used by the endpoint calculation, let

$$x(P) = \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{74}$$

At $P = P_{\mathrm{C}}$,

$$S_{\mathrm{required}} = 0.89540013264765879780\ldots \tag{75}$$

$$c_Q = 0.65802575992715543563\ldots \tag{76}$$

where

$$S_{\mathrm{required}} = 1 - x + c_Q x^2. \tag{77}$$

8 Hadronic transport and identifiability

8.1 Why Raw PDG $\Delta\alpha^{(5)}_{\rm had}(M_Z)$ Is a Different Quantity

The direct PDG/CERN diagnostic uses

$$A_Z(P_{\rm C}) = 128.30796547328624820996\ldots \tag{78}$$

$$\Delta_{\rm lep}(P_{\rm C}) = 4.30939786645220402713\ldots \tag{79}$$

$$\Delta\alpha^{(5)}_{\rm had}(M_Z) = 0.02761. \tag{80}$$

It forms

$$A_L = A_Z + \Delta_{\rm lep}, \qquad A_{\rm PDGdiag} = \frac{A_L}{1 - \Delta\alpha^{(5)}_{\rm had}(M_Z)}. \tag{81}$$

Numerically,

$$A_{\rm PDGdiag} = 136.38289507269557712141\ldots \tag{82}$$

This differs from 137.035999177. Holding this raw hadronic denominator shift fixed, the target would require

$$\Delta\alpha_{\rm had,req} = 0.03224434355788728882\ldots \tag{83}$$

or a same-scheme source-anchor bridge of

$$\begin{gathered}(\Delta\alpha_{\rm had,req} - \Delta\alpha^{(5)}_{\rm had}(M_Z)) \cdot \alpha^{-1}(0) = \\ 0.63507189998457776290\ldots\end{gathered} \tag{84}$$

inverse-alpha units, with $\Delta\alpha_{\rm had,req}$ from Eq. (83), $\Delta\alpha^{(5)}_{\rm had}(M_Z) = 0.02761$, and $\alpha^{-1}(0) = 137.035999177$ the CODATA central endpoint. With the pinned KNT19 payload $\Delta\alpha^{(5)}_{\rm had}(M_Z^2) = 0.027609 \pm 0.000112$, the same construction gives $\alpha^{-1}(0) = 136.3827548$, inside the interval from 136.3670481 to 136.3984652. The measured value lies outside this interval. The corresponding same-scheme anchor gap lies in $[0.6199, 0.6506]$ inverse-alpha units and contains the standard on-shell reference deficit 0.6309; both numbers measure the same anchor gap, computed once with the OPH lepton transport and once with standard running, so their agreement checks the bookkeeping and does not make the endpoint compatible with the measured value. This compare-only row does not identify the physical source of the gap [6].

8.2 The Measured Hadronic Spectral Input

The standard measured hadronic vacuum-polarization relation uses

$$\Delta\alpha_{\rm had}(q^2) = -\frac{\alpha q^2}{3\pi}\,\mathrm{P.V.}\int \frac{R(s)}{s(s-q^2)}\,\mathrm{d}s, \tag{85}$$

where $R(s)$ is the measured ratio of the bare $e^+e^- \to$ hadrons cross section to the pointlike $e^+e^- \to \mu^+\mu^-$ cross section.

The independently documented measured quantity in the standard electroweak literature is usually quoted as $\Delta\alpha^{(5)}_{\rm had}(M_Z^2)$. It is a dimensionless change in the running electromagnetic coupling at

the Z-boson mass, not the inverse-alpha correction $\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$ used in Eq. (68). For example, Davier, Hoecker, Malaescu, and Zhang report

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (275.7 \pm 1.0) \times 10^{-4}$$

from e^+e^--based data [11]. A perturbative update by Erler and Ferro-Hernandez gives

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (276.29 \pm 0.38 \pm 0.62) \times 10^{-4}$$

when low-energy cross-section data are used as input [14].

The comparison residual in Eq. (68) is different. It is expressed in inverse-alpha units and in the endpoint convention used by $A_Z(P)$. The standard published hadronic-running values document the physics source of the correction. They do not by themselves give the OPH same-scheme inverse-alpha correction to arbitrary precision. The community does publish the data machinery needed for this kind of calculation. Jegerlehner's alphaQED package provides hadronic running routines, covariance data, and an integration routine for custom kernels [16]. That is the right empirical route once the OPH endpoint kernel and finite same-scheme remainder are fixed.

In the OPH endpoint convention this is represented as a same-current spectral functional,

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s, \tag{86}$$

plus the same-scheme finite remainder needed by Eq. (72).

9 No-Go: The 24-Slot Register Scale Does Not Determine the Hadronic Residual

Theorem 1 (Coarse 24-slot register data do not determine the Thomson residual)**.** *The specified OPH source objects P, $\alpha_U(P)$, $N_c = 3$, $N_g = 3$, and the 24-slot repair-register count do not determine the exact low-energy Thomson residual.*

Proof. The hadronic endpoint contribution depends on the Ward-projected spectral functional

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int_0^\infty \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s \tag{87}$$

up to the same-scheme finite endpoint remainder. Let

$$K_P(s) = \frac{m_Z(P)^2}{3\pi s[s+m_Z(P)^2]}.$$

This kernel is not constant. Two positive spectral measures can therefore have the same coarse register count, the same leading normalization, and the same electroweak branch data while giving different residuals. For example, with $s_1 \neq s_2$,

$$\rho_1(s) = M\delta(s-s_1), \qquad \rho_2(s) = M\delta(s-s_2)$$

have equal total mass, but

$$\int K_P(s)\rho_1(s)\,\mathrm{d}s = MK_P(s_1), \qquad \int K_P(s)\rho_2(s)\,\mathrm{d}s = MK_P(s_2),$$

which are unequal because K_P is strictly decreasing on positive support. Thus the register count does not select the spectral measure or the residual. Even at fixed total mass M, moving a positive atom toward zero or toward infinity makes its moment arbitrarily large or small. A quantitative scale bound needs additional support or moment assumptions. □

The hadronic precision audit sharpens the no-go statement. The $\rho_Q(s;P)$ object used here is the two-current marginal of a larger source-derived hadronic spectral backend: a source QCD quotient ensemble, source parameter map, Ward-normalized current data, Stieltjes/Jacobi export, same-scheme remainder, and systematics/no-target-leak receipt bundle. That marginal can close the running-α/HVP transport only after it is emitted by the backend. It cannot be recycled as a full hadronic precision source, because HLbL and rare decay long-distance amplitudes require four-current and transition spectral data from the same source law.

Theorem 2 (Tautological 24-Jacobi realization)**.** *For any positive residual $R > 0$, there is an exact 24-point Jacobi/Stieltjes representation that reproduces R. Such a representation is non-predictive unless its nodes, weights, normalization, and finite remainder are emitted by OPH source rules before comparison with the Thomson endpoint.*

Proof. Fix $m > 0$. Choose pairwise distinct positive spectral nodes $s_1, \dots, s_{24} > 0$ and positive normalized weights $u_j > 0$ with $\sum_j u_j = 1$. The finite inverse Stieltjes/Favard construction gives a positive 24×24 Jacobi matrix $J_{24,Q}$ whose e_1-spectral measure is

$$\mathrm{d}\mu_0(s) = \sum_{j=1}^{24} u_j \delta(s - s_j).$$

Equivalently,

$$e_1^\top f(J_{24,Q}) e_1 = \sum_{j=1}^{24} u_j f(s_j)$$

for every function f on the spectrum. Define

$$D(J_{24,Q}; m) = e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1.$$

Since $s_j > 0$,

$$D(J_{24,Q}; m) = \sum_{j=1}^{24} u_j \left(\frac{1}{s_j} - \frac{1}{s_j + m^2} \right) > 0.$$

For any same-scheme finite remainder $\Xi_Q < R$, set

$$\omega_Q = \frac{3\pi(R - \Xi_Q)}{D(J_{24,Q}; m)}.$$

Then

$$\frac{\omega_Q}{3\pi} e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1 + \Xi_Q = R.$$

Distinctness makes the moment inner product nondegenerate through degree 23, so the finite Jacobi construction has 24 sites with positive off-diagonals. Repeated nodes instead collapse the support and do not justify a 24-site irreducible Jacobi matrix. The construction reproduces the chosen residual exactly, but the freedom in s_j, u_j, ω_Q, and Ξ_Q means it is a back-solving theorem rather than a source derivation. □

The source-side object needed for a final OPH endpoint theorem is therefore

$$R_Q(P) = \frac{\omega_Q(P)}{3\pi} e_1^\top \left[J_{24,Q}(P)^{-1} - \left(J_{24,Q}(P) + m_Z(P)^2 I \right)^{-1} \right] e_1 + \Xi_Q(P), \tag{88}$$

or equivalently the source-emitted spectral density $\rho_Q(s;P)$ plus the same-scheme finite remainder. The required dependency graph has no path from the measured Thomson endpoint into $J_{24,Q}$, ω_Q, Ξ_Q, or ρ_Q. The available execution embeds target constants, uses a mismatched P, conflates the closure residual with the total payload, and supplies a sampled envelope rather than a certified interval. It does not support a physical conclusion. A qualifying contract fixes one coordinate schema, requires a source-emitted function over the same P, certifies quadrature and interval bounds, and scores a detached artifact. The local source-payload validator checks declared contract consistency; it does not replay the named backend proof or establish spectral provenance. An arbitrary nonempty certificate label cannot establish those premises; in particular, a payload whose anchor and other components sum to an interval disjoint from its declared total endpoint is inconsistent. Such inputs are rejected, and a consistent but unreplayed contract is non-promoting. The separate executable interval certificates for the two incomplete declared maps retain their stated mathematical scope.

10 Pixel feedback with the memory mass law

The transport maps above use a supplied phenomenological charged spectrum. A common feedback loop instead evaluates one charged-fermion kernel on the mass coefficients that the memory construction realizes. Keep Eqs. (12), (15), (19), (21) and the heat equation (29) with infinite representation sums. Let $u = \alpha_U$, let E_* be a declared energy unit in the role of E_P, and write β for the transmutation coefficient β_{EW}. The charged leptons use the conditional minimal complete public response law

$$\kappa = \frac{u/50}{1 + u/31 + u^2/310}, \quad \chi = \frac{u^2/512}{1 - u/77}, \quad \zeta = \frac{\alpha_P^2/21}{1 - u/27 - u^2/135},$$
$$\alpha_P = \frac{P - \varphi}{\sqrt{\pi}}, \quad s_j = \Big[1 + \sqrt{2}\, \mathrm{e}^{-\chi} \cos\Big(\frac{2}{9} + \zeta + \frac{2\pi j}{3}\Big)\Big]^2, \quad t_j = s_j \Big[\frac{\mathrm{e}^{3\kappa}}{2 \cdot 6^{14} \prod_k s_k}\Big]^{1/3},$$

with masses $m_j = v t_j$. The quark coefficients are $(6^{-6}, 6^{-3}, 1)/\sqrt{2}$ and $(6^{-6}, 6^{-4}, 6^{-2})/\sqrt{2}$ in units of v. With K_f from Eq. (42) at $Q^2 = m_Z^2$, for all three charged leptons and all six quarks including the top, the response is

$$A = \alpha_2^{-1} + \frac{5}{3}\alpha_1^{-1} + \sum_\ell K_\ell + \Big(1 - \frac{3\alpha_3}{\pi}\Big) \sum_q K_q, \qquad \alpha \mapsto P = \varphi + \sqrt{\pi}\, \alpha \mapsto (u, m_Z, v, t_j) \mapsto A \mapsto \frac{1}{A}.$$

The quark factor is an inherited prescription. The mass architecture has target-informed provenance, with constants $50, 31, 310, 512, 77, 21, 27, 135$, while its execution reads no measured mass, measured coupling, GeV calibration or stored pixel root.

Theorem 3 (Two certified feedback designs)**.** *For the two designs in Table 2 the loop has a locally unique fixed point. Outward interval arithmetic with enclosed infinite heat tails gives a strict self-map on an α interval of radius 10^{-14} with $|g'| < 0.00734$, so Banach's theorem gives one root in that interval. Every running coupling and response denominator is positive on the enclosure, and both native-memory inverse branches are admissible throughout it.*

Design	$(b_1, b_2, b_3),\ \beta$	P	α^{-1}	u
Supersymmetric proxy	$(33/5,\ 1,\ -3),\ 4$	1.630971514159316	137.000994766342	0.041124260725897
Minimal census	$(41/10,\ -19/6,\ -7),\ 7$	1.628533369611833	168.815082928453	0.019531387695090

Table 2: Certified conditional roots of the feedback loop. The ratio v/E_* is $2.01983668924449\times10^{-17}$ for the first design and $8.61718526720080 \times 10^{-21}$ for the second.

An independent replay solves the three simultaneous equations in $(1/u, \log(M_U/m_Z), \alpha)$ with direct quadrature of each fermion integral. The minimal coefficients are exact traces of the native fifteen-state internal currents with the intended chiral multiplicities. The first design requires a supplied running proxy whose extra fields are absent from the native matter carrier; its value differs from the measured endpoint by about 0.0350. The second uses the native census and sets β to the magnitude of its color coefficient, a choice that no theorem identifies with the Higgs scale.

Proposition 2 (Minimal census with coefficient four)**.** *With $b = (41/10, -19/6, -7)$, $\beta = 4$, $P \geq 1$, $M_U \geq m_Z$, positive running couplings and no intermediate thresholds, the heat and scale equations have no solution. This excludes that combined census and scale prescription, and leaves other scale laws, intermediate thresholds and other censuses outside its statement.*

The proof is in the particle construction [2]: the scale law forces $\alpha_2^{-1} < 6$ and $\alpha_3^{-1} \leq \alpha_2^{-1}$, so the combined left side of Eq. (29) lies below $1/80$, whereas closure requires at least $P/4 \geq 1/4$. The same construction gives the memory attachment: on each enclosure both inverse controllers obey $0 < p < 1/12$ and $0 < q < 1/6$ and feed the complete-record mass operator. The kernel supplies no charged-vector or charged-scalar terms, supersymmetric threshold matching, confining hadronic spectral measure, pole-mass convention or native-current derivation of the Thomson limit, and the particle paper's comparison table derives no measured mass, so neither root is a physical value of α.

11 The conditional physical theorem

11.1 The Conditional Pure Source Theorem

Theorem 4 (OPH fine-structure endpoint, conditional pure source form)**.** *Assume:*

(i) the OPH overlap-consistency branch emits the source map $P \mapsto A_Z(P)$ by Eqs. (12)–(32);

(ii) the Ward-projected $\mathrm{U}(1)_Q$ transport theorem emits a same-scheme endpoint map

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P);$$

(iii) a target-independent source law selects this source and transport map from the admitted map family;

(iv) a typed same-quantity bridge identifies $(P-\varphi)/\sqrt{\pi}$ and $1/A_{\mathrm{Th}}(P)$ as readings of one physical screen cell, with $1/A_{\mathrm{Th}}(P)$ identified as its Thomson-limit electromagnetic coupling;

(v) $G(P) = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ is a self-map and a contraction on a nonempty closed branch interval $I \subset \mathbb{R}$;

(vi) the interval image contains the root and the residual bound is certified.

Then there is a unique $P_\star \in I$ satisfying $G(P_\star) = P_\star$, and the fine-structure constant on that branch is

$$\alpha(0) = \frac{1}{A_{\mathrm{Th}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

Proof. By assumption (v), I is nonempty and complete in the Euclidean metric, and $G : I \to I$ is a contraction. Banach's fixed-point theorem gives a unique $P_\star \in I$ such that $G(P_\star) = P_\star$. Equation (11) gives

$$P_\star - \varphi = \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P_\star)}.$$

Dividing by $\sqrt{\pi}$ gives

$$\frac{P_\star - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P_\star)}.$$

The left side is $\alpha_{\mathrm{ext}}(P_\star)$, and the right side is $\alpha_{\mathrm{in}}(P_\star)$. Assumptions (iii)–(iv) select this map and identify their common value with the Thomson-limit electromagnetic coupling $\alpha(0)$. □

The interval form of assumptions (v)–(vi) is direct: on any nonempty closed interval I where interval-arithmetic evaluation certifies $G(I) \subseteq I$ together with a derivative bound $|G'| \le L < 1$, the Banach fixed-point theorem gives both existence and uniqueness of the fixed point in I. The public interval contraction certificate [3] proves $G(I) \subseteq \mathrm{int}(I)$ and $L \le 0.0724$ for both readout modes by a direct Banach argument in mean-value (centered) form, with `mpmath.iv` outward rounding on every elementary operation and with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants, so the enclosure covers the infinite-cutoff sums. The certified unique fixed points are $\alpha^{-1} = 136.994835177413\ldots$ for the source map (enclosure width 7.2×10^{-24}) and $\alpha^{-1} = 137.035660136946577\ldots$ for the gauge-width map. The certificate certifies the declared numerical map; it does not relate either fixed point to the measured constant. Both roots lie outside the measured comparison basin. The global at-most-one statement is discharged on the declared analytic domain ($\alpha^{-1} \in [100, 200]$): $\sup|g'| < 1$ on all 256 certified pieces, both readout maps, zero exceptional set (domain-global uniqueness certificate in the same artifact bundle), so each map has exactly one fixed point on the domain. A maximal-domain extension certificate closes the exterior. An envelope lemma built from the declared solver windows (the pixel scan window $\alpha_U \in [0.02, 0.08]$ and the m_Z log-grid window $\ln(\mu_U/m_Z) \in [0, 50]$) together with the m_Z-closure identity certifies $1/\alpha_3(m_Z) \ge 4$ and a positive inverse-alpha readout floor on the whole analytic domain, and an interval sweep of the full pixel window certifies that every window-consistent fixed-point candidate of either readout map lies inside the declared solver interval (global uniqueness extension certificate). Each declared map therefore has exactly one fixed point on its maximal analytic domain, and the set of fixed points outside the declared interval is empty; the headline enclosures above are unchanged.

A source-only theorem requires a source-derived Ward-current spectral measure, the source Jacobi kernel, electromagnetic normalization, a same-scheme finite remainder, a target-independent rule selecting one physical map, a typed construction that identifies its two sides as the same quantity, a dependency graph excluding Thomson-target leakage, a full contraction interval for the selected map, and a prediction interval narrower than the adopted direct measurement. For the comparison standard used here, the direct rubidium recoil result of Morel, Yao, Cladé, and Guellati-Khélifa

reports $\alpha^{-1} = 137.035999206(11)$ with 81 parts-per-trillion relative precision [13]. The operational receipt is therefore

$$\mathrm{rad}(\alpha_{\mathrm{pred}}^{-1}) < 1.1 \times 10^{-8}$$

in inverse-alpha units for the cited direct rubidium recoil interval.

12 An independent entangled-pair spectral candidate

Wissner-Gross gives an exact algebraic candidate for the Thomson coupling from a finite entangled-pair construction [20]. Its scalar can be checked independently of the proposed physical identification. For four slots of dimensions $(3, 2, 3, 2)$, traceless letters on unordered slot pairs give 169 product records. Three additional exchange-odd directions give a 172-dimensional nonzero feature space. The two commuting signed exchanges X, Y satisfy $X^2 = Y^2 = I$. For

$$K = \frac{(2I - X)(3I - Y)}{2},$$

the joint projectors have gains $1, 2, 3, 6$ and ranks $79, 48, 42, 3$. The complete declared feature normalization is $(3^2 - 1)(2^2 - 1)(3 \cdot 2)^4 = 31104$. Thus

$$\alpha_{\mathrm{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\mathrm{pair}}^{-1} = 137.035999189566155\ldots.$$

The executable reconstruction checks the involutions, integer projector identities, ranks and gains. It fits no continuous coefficient. These are properties of the stated finite operator and readout normalization.

The six-dimensional matrix algebra and a 24-dimensional mixed traceless space are compatible finite ingredients for comparison with OPH. Equal dimensions alone do not identify a state, a current representation or a photon two-point function. Matching invariance and reading the identity as one also do not uniquely select normalized trace: the normalized Bell projector state and the tracial state share those properties and evaluate the Bell projector differently. The normalized trace is therefore retained as an explicit readout choice in this comparison.

A complete OPH instrument retains the complementary Bell outcome. For two flat 3×2 pairs and the canonical qutrit Bell projection, its success probability is $1/6$, with failure probability $5/6$. Independently embedded flat supports need not contract to a flat pair: a tested pair of flat factors gives reduced eigenvalues $1/5, 4/5$ after contraction. Aligned supports or an adjusted contraction avoid that example, and are additional hypotheses. These controls bound the reuse of the pair construction without changing its canonical scalar arithmetic.

If the independent same-cell identification $P = \varphi + \sqrt{\pi}\,\alpha(0)$ is granted and $\alpha(0) = \alpha_{\mathrm{pair}}$, substitution gives

$$P_{\mathrm{pair}} = 1.630968209402773261\ldots.$$

This is a conditional coordinate, not a fixed-point theorem for either trial map analyzed in this paper. The two interval-certified trial-map roots retain their own hypotheses and endpoint tests. Substituting another coupling value does not modify those maps or supply their missing transport.

The same conditional coordinate can be checked across scales without a new fit. Exact rational square-root enclosures and alternating-series bounds for π reproduce $\alpha_{\mathrm{pair}}^{-1}$ and propagate it through the declared relation $n_s = 1 - P/48$. Relative to the CODATA 2022 inverse-alpha

reference, the scalar differs by approximately 0.5984 reported standard uncertainties. Its tilt is $0.966021495637442\ldots$, only 2.47×10^{-14} above the CODATA-substituted tilt. Both ends of the rational enclosure give the same weighted lower-tail count 3114/508985 in the retained Planck+ACT DR6+lensing+DESI BAO posterior projection. This count is a posterior diagnostic, not a frequentist significance; the numerical enclosures bound arithmetic error, not physical model error. The comparison therefore leaves that conditional cosmological result unchanged. The independent calculation and the joint-construction scope check are part of the executable model's empirical diagnostics [10]. Neither a close coupling value nor a separately supplied oscillator or Weyl–Higgs action identifies one physical model across these measurements.

For a physical coupling claim, the same observer construction must identify the conserved electromagnetic current, its charge normalization, a massless photon pole, and the appropriate dressed-to-bare residue or equivalent low-momentum response, with controlled running and thresholds. The finite candidate supplies no such OPH attachment. Its numerical proximity to the experimental value is a retrospective comparison and is not a prospective prediction record. The calculation and its controls are self-contained in the executable model [10]; its elementary normalization and pixel-substitution algebra is formalized in the repository's Lean library.

12.1 A conditional combined gauge branch

The exact matter census also supplies representation indices $k = (10/3, 2, 2)$. If this column is identified with the physical kinetic ray, and the three-family, one-doublet Standard Model one-loop column is imported, write

$$(x_Y, x_2, x_3) = c(10/3, 2, 2) + t(41/6, -19/6, -7), \qquad x_i = \alpha_i^{-1}.$$

Adding the declared same-scale tree relation $x_Y + x_2 = \alpha_{\text{pair}}^{-1}$ and representation-entropy relation $\ell_2(\alpha_2) + \ell_3(\alpha_3) = P_{\text{pair}}/4$ gives an isolated positive branch. The entropy functions here use the stated representation dimensions and quadratic Casimirs with weights proportional to $d_R \exp(-4\pi^2 \alpha C_2(R))$. A monotonicity and tail certificate encloses the unique root in

$$5.002666247 < t < 5.002666250.$$

Its dimensionless couplings are approximately $\alpha_Y = 0.00922782$, $\alpha_2 = 0.03488203$, and $\alpha_3 = 0.10536109$. The electroweak quadratic mass form then has one photon null direction and charged-to-neutral ratio 0.79079908, per unspecified vacuum norm. These values consume the declared kinetic ray, one-loop census, entropy law and same-scale matching. They are not measured Z-scale predictions: Thomson-to-running transport, thresholds, physical kinetic normalization, vacuum scale and pole matching are not supplied. The executable coefficient and mass-form calculations retain those inputs explicitly.

Relation to calibrated matter sources. A positive native-current hadron action can fit proton, neutron and charged-pion energies and an axial matrix element with four empirical inputs. Its semileptonic extension has an exact electron threshold and bounded nucleon energy records [2]. Those finite calibrations do not determine the electromagnetic spectral measure required here. In particular, substituting fitted composite energies for thresholds in the charged-fermion return map does not transport the current, reference state or finite matching terms. The phenomenological screening controls and the complete calibrated semileptonic source therefore retain distinct domains; neither supplies a measured Thomson coupling through numerical mass agreement alone.

13 Reproducibility

The executable fixed-point calculation is available in the public repository [3]. From the code directory cited there, the main human-facing CLI is

```
python3 derive_p.py --color always
```

To print only the pure source value:

```
python3 derive_p.py --no-hadron-closure
```

To emit machine-readable output:

```
python3 derive_p.py --json --output runtime/report.json
```

To run the raw PDG/CERN diagnostic discussed in Section 8.1:

```
python3 fine_structure_fixed_point_demo.py --compare-alpha-inv 137.035999177
```

14 Scope of a physical conclusion

The equation $P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ supplies a fixed-point test for a specified return map. The mathematical certificates distinguish its existence and uniqueness from identification of A_{Th} with the physical Thomson response. A physical result requires the same current, charged spectrum and interaction to generate the Ward-projected spectral measure, with consistent thresholds, finite matching, clock units and a same-cell map. The measured comparison coordinate P_{C} and the back-solved hadronic residual are calibrations. Neither is an independent output of that source construction.

Declarations

Funding declaration. No funding was received for this work.

Consent to Participate declaration. Not applicable.

Consent to Publish declaration. Not applicable.

Author Contribution declaration. Bernhard Müller conceived the OPH fine-structure fixed-point formulation, prepared the mathematical derivation, wrote and checked the numerical calculation, wrote the manuscript, and prepared the reproducibility materials. Alexander Osika contributed conceptual framing, manuscript review, and approval of the final manuscript.

Data Availability declaration. No experimental data were generated. The manuscript cites the 2022 CODATA/NIST inverse fine-structure constant [17, 18], public hadronic cross-section and vacuum-polarization references [15, 11, 12], and the OPH source materials cited below. The TeX source, OPH paper sources, and fixed-point calculation code are available in the public repository [3].

Code Availability declaration. The fixed-point code and runtime artifacts are available in the public repository [3].

Ethics declaration. Not applicable.

Competing Interests declaration. The authors declare no competing interests.

References

[1] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf.

[2] B. Müller, M. Poneder, A. Osika, K. Xue, and D. A. Friedman, *Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/paper/deriving_the_particle_zoo_from_observer_consistency.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[3] OPH fixed-point calculation code, https://github.com/FloatingPragma/observer-patch-holography/tree/r2066/code/P_derivation. Primary files: `paper_math.py`, `derive_p.py`, `fine_structure_fixed_point_demo.py`.

[4] B. Mueller and B. de La Fournière, *The Positive-Chamber Koide Identity for Icosahedral Face Circulants*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/extra/koide_identity_from_positive_c3_face_circulants.pdf.

[5] FloatingPragma, *OPH predictions and public-data comparisons*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md.

[6] FloatingPragma, *Postdiction ledger*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/code/particles/runs/status/postdiction_ledger.json.

[7] FloatingPragma, *The frozen-prediction register*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/claims/frozen_prediction_register.json.

[8] FloatingPragma, *P-dependent electromagnetic observation and native feedback*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/model/observer/p_response/README.md.

[9] FloatingPragma, *Tangent first laws and finite entropy changes: native reference entropy and region boundaries*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2066/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md.

[10] FloatingPragma, *The runnable OPH model*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2066/model.

[11] M. Davier, A. Hoecker, B. Malaescu, and Z. Zhang, *Reevaluation of the Hadronic Contributions to the Muon $g-2$ and to $\alpha(M_Z)$*, Eur. Phys. J. C **71**, 1515 (2011). https://arxiv.org/abs/1010.4180.

[12] A. Keshavarzi, D. Nomura, and T. Teubner, *The $g-2$ of charged leptons, $\alpha(M_Z^2)$ and the hyperfine splitting of muonium*, Phys. Rev. D **101**, 014029 (2020). https://arxiv.org/abs/1911.00367.

[13] L. Morel, Z. Yao, P. Cladé, and S. Guellati-Khélifa, *Determination of the fine-structure constant with an accuracy of 81 parts per trillion*, Nature **588**, 61–65 (2020). https://doi.org/10.1038/s41586-020-2964-7.

[14] J. Erler and R. Ferro-Hernandez, *Perturbative contributions to $\Delta\alpha^{(5)}(M_Z^2)$*. https://arxiv.org/abs/2308.05740.

[15] Particle Data Group, *Data files and plots of cross-sections and related quantities in the Review of Particle Physics*. https://pdg.lbl.gov/current/xsect/.

[16] F. Jegerlehner, *Software packages alphaQED and pQCDAdler*. https://people.physik.hu-berlin.de/~fjeger/software.html.

[17] NIST, *CODATA Value: inverse fine-structure constant*, 2022 CODATA recommended value. https://physics.nist.gov/cgi-bin/cuu/Value?eqalphinv=.

[18] P. J. Mohr, E. Tiesinga, D. B. Newell, and B. N. Taylor, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*. NIST publication page: https://www.nist.gov/publications/codata-recommended-values-fundamental-physical-constants-2022.

[19] NIST/CODATA, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*, wall chart. https://physics.nist.gov/cuu/pdf/wall_2022.pdf.

[20] A. D. Wissner-Gross, *A Parameter-Free Candidate for the Fine-Structure Constant from Entangled Qudits*, preprint (2026). https://alexwg.org/publications/ChainTheoryAlpha.pdf.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.